

1. (a) Large k with noisy data
2. (b) Small changes in data lead to different trees.
3. (c) Reducing Variance
4. (c) All features are considered at each split
5. (d) perfect multicollinearity exists,
6. (c) Sigmoid
7. (c) Accuracy
8. (c) Training time
9. (b) To improve accuracy always.
10. (c) Logistic Regression

Ans $\Rightarrow$ Overfitting in Decision tree.
Overfitting occurs ~~due to~~ when the model performs better in the training data but "poor" in testing dataset.

The overfitting in decision tree occurs ~~because~~ when the depth of the tree is very large, when it tries to cover all data points of the dataset.

We can reduce overfitting by limiting the depth of the tree.

Bagging and Random forest address the problem of overfitting.

Bagging is an ensemble technique in which a dataset is divided into many bootstrap samples and then these samples are passed to different models and their predictions are aggregated using majority voting for classification, averaging for regression.

It uses multiple models to find output.

Random forest is also an Ensemble technique which uses

- Multiple decision trees.
- Random feature selection of bootstrap samples to predict the output.
- It uses bagging approach in which multiple bootstrap samples with random feature selection are passed through multiple decision trees and predicts the output.

Ans ⇒ Working of Random Forest.

1. Bootstrap Sampling,

In bootstrap sampling, ~~the~~ the dataset is divided into multiple dataset with replacement or without replacement. Here, the multiple datasets contains data that is created by replacement (not overplacement).

Random feature Selection.

After creating the bootstrap sample, the random feature selection are performed used to reduce overfitting by introducing randomness in the data.

Majority Voting

After feature selection, the data is passed through multiple decision tree and predictions are made. This prediction are aggregating using

Majority Voting for Classification
Averaging for Regression.

Ans:

	Predicted fraud	Predicted not fraud
Actual Fraud	120 (TP)	30 (FN)
Actual Not fraud	50 (FP)	800 (TN)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{120 + 800}{120 + 800 + 30 + 50} = \frac{920}{1000}$$

$$\text{Accuracy} = 92\%$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{120}{120+50} = \frac{120}{170} = 0.70 = 70\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{120}{120+30} = \frac{120}{150} = 80\%$$

$$\text{F1 Score} = \frac{2 \times P \times R}{P+R} = \frac{2 \times 0.7 \times 0.8}{0.7+0.8} = \frac{1.12}{1.5} = 0.746 = 74.6\%$$

No this model is not acceptable for fraud detection because their recall is 80%. means out of 100 frauds, it can detect only 80 correctly.

Ans

120 max-feature = None

It considers all the features at each split and causes high training time and computation cost causes overfitting.

⇒ Random forest overfit less than a single decision tree because

- it uses multiple decision trees to make prediction
- It uses random feature selection to introduce randomness in the data.

Single decision tree is more overfit due to

- Sensitive to change in dataset (high variance).

⇒ $n_{\text{estimators}} \approx 200$

does not impact on Variance or bias.

It increases the training time. It does affect the Variance or bias.